

hiptron

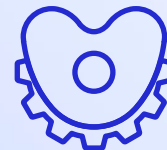


COMPANY NAME

hiptron is an early-stage startup from Aachen developing a **smart sensor module for existing rollators**. The system combines hardware and software to **improve everyday mobility, safety and independence for older adults** while also giving relatives more orientation and reassurance.

Founded in 2025, hiptron currently consists of a **two-person founding team** supported by 2–3 additional contributors across mechanical engineering, software development and digital product development.

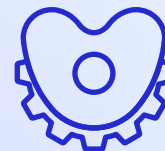
The **business model** combines a **one-time hardware sale** for the sensor module with **recurring subscription services** for connectivity, mobility insights and family features



CHALLENGE TITLE & GUIDING QUESTION

Understanding Mobility Changes Before Emergencies Happen

How can we use AI to detect and explain meaningful changes in everyday mobility behaviour before critical events occur in a way that feels supportive rather than surveilling?



CHALLENGE DESCRIPTION

PROBLEM

Mobility decline happens gradually through small behavioural changes like shorter walking distances, fewer outdoor activities or reduced routines. Current solutions focus on emergencies or tracking not on understanding everyday mobility.

TARGET GROUP

- Older adults using rollators
- Relatives seeking reassurance without surveillance

Main user perspective for this challenge:

Relatives trying to understand meaningful mobility changes over time.

STATUS QUO

Existing solutions often:

- react too late
- feel intrusive or medical
- provide little understandable insight
- lack trust and explainability

The challenge is to translate mobility behaviour into insights.

EXAMPLES

- Shorter walking distances over weeks
- Fewer outdoor activities
- More frequent stops or pauses
- Shrinking daily activity radius

SCOPE

In Scope:

- AI mobility pattern analysis/ insights
- Dashboard & UX concepts
- Behaviour trend interpretation

Out of Scope:

- Medical diagnosis
- Emergency/fall detection systems
- Production-ready software



CARE	4	SUSTAINABLE INNOVATION

CHALLENGE SUMMARY

hiptron develops a smart sensor module for rollators to better understand everyday mobility changes in older adults. The challenge is to explore:

How can **AI help identify and explain meaningful changes in everyday mobility behaviour** before critical events occur?

How can **mobility patterns be translated into understandable, trust-based insights for relatives and users** without creating a feeling of surveillance?





ADDITIONAL INFORMATION & MATERIALS

Students will receive:

- Current hiptron product concept and challenge brief
- Draft system architecture: sensor module → connectivity → cloud → app
- Initial app role logic for older adults and relatives
- Insights from user interviews with seniors, relatives and care-related stakeholders
- Examples of potential mobility signals: distance, routines, pauses, outdoor activity, activity radius

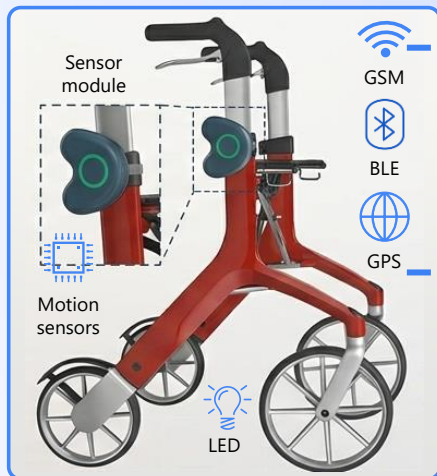
Important context:

- hiptron is positioned as a non-medical safety and mobility product
- The goal is not diagnosis, surveillance or fall detection
- The key challenge is translating mobility data into simple, trustworthy and human-centered insights
- Privacy, trust and emotional acceptance are as important as technical accuracy

ADDITIONAL INFORMATION & MATERIALS

Hipton System Architecture

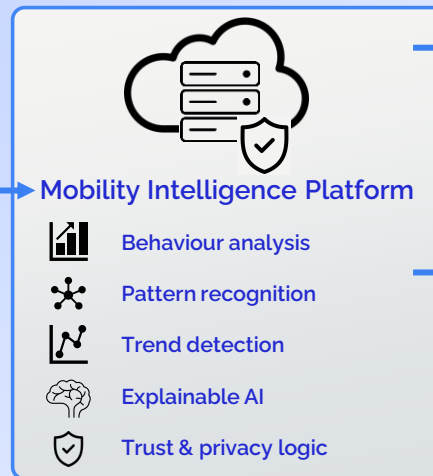
Rollator + Sensor Module



Connectivity Layer



Cloud & Intelligence



App Layer

